

Chemical Engineering Materials ChE 350

Unique # 16060; Spring 2025

Class Time: 10:00 – 11:00 a.m. on Mondays, Wednesdays, and Fridays

Class location: CPE 2.216

Recitation: 8:00 – 9:15 a.m. on Tuesdays

Recitation location: CPE 2.216

Instructor: Joaquin Resasco (he/him/his)

Student Hours: M 11:00-12:00 p.m. EER 5.702

Email: resasco@utexas.edu

TA: Allison Segapeli (she/her/hers)

Homework help session: W 3:00-5:00 p.m. EER 6.702

Email: ajsegapeli@utexas.edu

What will I learn? This course introduces the fundamentals of materials science and engineering. In this class, you will learn why materials behave the way they do by understanding their structure. You will learn how materials respond to applied stimuli such as mechanical forces, electricity, and light and link this to their atomic, crystal, and microstructure. With this understanding, you will see how we can design new materials with useful properties.

Textbook: W. D. Callister & D. G. Rethwisch, Materials Science and Engineering: An Introduction, 10th Edition; ISBN: 978-1-119-40549-8 (John Wiley & Sons, New York, NY).

A PDF of the course textbook is available on Canvas.

Course requirements: The coursework will consist of reading, problem sets, in-class quizzes, three midterms, and a final exam. Please bring a laptop or cell phone with the Canvas student app installed to class. We will need this for quizzes and polls.

Grading: **Problem sets:** (10%) Due Friday 9:59 a.m. (before start of class). Problem sets will be uploaded online to Canvas. Lowest grade dropped.

Quizzes: (20%) Mondays 10:00 a.m. (start of class) on Canvas. Two dropped.

Midterm exams 1,2,3: (50%, 16.7% each) During recitation (Tuesdays 8:00 9:00 a.m.) in CPE 2.216

Final exam: (20%): May 1st 8:00-10:00 a.m. in CPE 2.216

Grades: >89% (A); 86-89% (A-); 82-86% (B+); 75-82% (B); 71-75% (B-); 67-71% (C+); 59-67% (C); 55-59% (C-); 49-55% (D); ≤ 49% (F)

Course outline

Topic	Associated reading
Introduction: The role of materials in our life	Callister 1.1-1.3
Classification of materials	Callister 1.4-1.6
Atomic structure and bonding	Callister Chapter 2
Crystal structure	Callister 3.1-3.7
Crystallographic positions, directions, and planes	Callister 3.8-3.12
X-ray diffraction	Callister 3.16
Defects in crystals	Callister 4.1-4.8
Nucleation	Callister 10.1-10.3
Diffusion in solids	Callister Chapter 5
Midterm 1 (Tuesday February 11th in recitation)	
Mechanical properties of metals	Callister 6.1-6.10
Isomorphous alloys	Callister 9.1-9.10
Eutectic alloys	Callister 9.11-9.14
Structure and properties of steel	Callister 9.18-9.20, 10.7-10.9
Structure and properties of ceramics	Callister 12.1-12.3, 12.5-12.7
Midterm 2 (Tuesday March 11th in recitation)	
Structure and properties of carbon materials	Callister 12.4
Structure and properties of polymers	Callister 14.1-14.12, 15.1-15.18
Optoelectronic properties of materials	Callister 18.1-18.15
Midterm 3 (April 15th in recitation)	
Materials and the environment	
Final exam (Thursday May 1st 8:00-10:00 a.m.)	

Note: the course schedule is tentative and subject to change. A lecture by lecture schedule is available on the course Canvas page.

How can I succeed in this class?

- **Attend class and participate!** Class is an opportunity to not just to listen to lectures but to practice applying material and to ask questions. Engagement in class will influence how you do on quizzes and exams.
- **Read the textbook before coming to class.** The textbook is readable and clearly laid out. We will follow it closely.
- **Stay up to date with the material.** This course covers a lot of ground, so try not to get behind.
- **Check Canvas regularly.** This will be our main communication tool throughout the course.
- **Get excited about materials!**

Class Policies:

COVID-19: COVID 19 has impacted many in our University of Texas at Austin community. With this in mind, this semester I commit to being adaptable in this difficult time. The course policies regarding attendance, grading, and assignments/exams are designed with this in mind. If you experience any hardships such as illness or family crisis, please know that these policies may be amended and therefore you should communicate with me as soon as you feel comfortable doing so. If for any reason you do not feel comfortable discussing with me, please visit Student Emergency Services. For additional campus resources, please visit <https://coronavirus.utexas.edu/students>.

Late work: Late homework **will be penalized at a rate of 0.5% for each minute late** unless a prior arrangement has been made with the instructor. The intent of this is to ensure that submission issues do not result in a zero on an assignment. Additionally, one homework and two quizzes will be dropped to account for unexpected circumstances.

Student hours: This time is set aside for all of you (hence the name student hours instead of office hours). This time can be used however you would like it to be. We can discuss the course, your experience at UT, a cool science topic, research, career plans, etc. If you would like to talk at another time, just let me know and we can find a time that works for you.

Video materials: In addition to our lectures, I will post material as short videos to Canvas. These should supplement the course material to explain important concepts or provide additional practice on solving problems. Concepts and ideas from these materials will appear on weekly quizzes.

Student feedback: I am committed to making this course an effective learning experience for everyone. Periodically we will have course surveys so you can tell me what is working and ways that I can improve the course.

Academic Integrity: You are expected to uphold and reflect the University Code of Conduct and the Student Honor Code in this course:

Student Honor Code: "As a student of The University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity."

University Code of Conduct: "The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility."

Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community.”

Cheating and plagiarism are not only counter-productive to learning the material but also carry serious consequences (failure of the course and/or dismissal from the University for academic dishonesty). Cheating includes communicating questions and answers to other students on quizzes and exams and using disallowed resources (cell phones, book, notes, etc.) to answer questions on an exam or assignment. Plagiarism includes copying another’s work and representing it as one’s own. It is your responsibility to know what plagiarism is, and how to avoid it: <http://www.lib.utexas.edu/services/instruction/learningmodules/plagiarism/> Cases of cheating and plagiarism will be referred to the Honor Code Council. You can find out more about academic integrity at: <http://deanofstudents.utexas.edu/sjs/scholdis.php>

Disabilities: I am committed to providing an accessible and inclusive learning environment in this course. If you need accommodations (or think you might) to perform your best, please talk with me **at the beginning of the semester**. Upon request, the University of Texas at Austin provides appropriate academic accommodations for qualified students with disabilities. For more information, contact the Services for Students with Disabilities: Call 512-471-6259, Videophone 512-410-6644, Email ssd@austin.utexas.edu, Website <http://ddce.utexas.edu/disability/>

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, slides, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University’s Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.

Class Recordings: Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

Counseling and Mental Health Center: There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. If you or anyone you know experiences any academic stress, difficult life events, or feelings like anxiety or depression, we strongly encourage you to seek support. <http://www.cmhc.utexas.edu/individualcounseling.html>

Title IX Reporting: Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence, and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working environment free from discrimination in all forms. When sexual misconduct occurs in our community, the university can: 1. Intervene to prevent harmful behavior from continuing or escalating. Provide support to students and employees who have experienced harm or have become involved in a Title IX investigation. 3. Investigate and discipline violators of the university's relevant policies.

Faculty members and certain staff members are considered "Responsible Employees" or "Mandatory Reporters," which means that they are required to report violations of Title IX to the Title IX Coordinator. I am a Responsible Employee and must report any Title IX related incidents that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a responsible employee. If you want to speak with someone for support or remedies without making an official report to the university, email advocate@austin.utexas.edu. For more information about reporting options and resources, visit titleix.utexas.edu or contact the Title IX Office at titleix@austin.utexas.edu.